CHANNEL 03 · ACTIVE CH-03 · 2.4 GHz



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CLASSIFIED — PROBLEM STATEMENT CHANNEL

HEALTHCARE & ASSISTIVE TECHNOLOGY

PROBLEM STATEMENTS · ROUND 1

24-HOUR NATIONAL-LEVEL HARDWARE HACKATHON

29–30 JANUARY 2027 · FR. C. RODRIGUES INSTITUTE OF TECHNOLOGY, VASHI

DOMAIN SLOT 03 CH-03 · 2.4 GHz



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HEALTHCARE & ASSISTIVE TECHNOLOGY

Elevating patient outcomes, clinical efficiency and physical independence through intelligent biomedical hardware and point-of-care diagnostics. Non-invasive, dependable devices that save critical seconds.

— AREAS TO EXPLORE

- Vital Signs Telemetry
- Rehabilitation Hardware
- Smart Emergency Triage
- Portable Diagnostics
- Sensory Feedback

— WHO SHOULD EXPLORE

Biomedical Instrumentation · Analog Sensor Integration · DSP · Wearable Tech & Health Informatics

> ***"The pulse of precision engineering — design the devices that give clinicians back their critical seconds."***

— IN THIS CHANNEL

PS-01	Continuous Temperature Monitoring & Excursion Alerts for Medical Cold Storage
PS-02	Indoor Navigation & Obstacle Detection Aid for Visually Impaired People
PS-03	Backup Power Monitoring & Failure Alert for Critical Hospital Equipment
PS-04	Real-Time Location & Availability Monitoring of Shared Hospital Equipment
PS-05	Affordable Hand Rehabilitation Device with Movement Assistance & Feedback

PS-01 CONTINUOUS TEMPERATURE MONITORING & EXCURSION ALERTS FOR MEDICAL COLD STORAGE

Recommended for First & Second Year students

A · PROBLEM DEFINITION

Vaccines, blood products and many medicines must be stored within a narrow temperature range (for example 2–8 °C for most vaccines). Refrigerators can drift out of range because of door openings, compressor faults or power cuts, and staff may not notice. The challenge is to monitor storage temperature continuously, detect excursions and alert staff quickly, while keeping a trustworthy record.

B · CONTEXT

Vaccines and blood products can lose potency, or become unsafe, when exposed to temperatures outside their range, and the damage is not visible. Many clinics, especially in rural areas, still depend on manual thermometer readings once or twice a day, which miss overnight failures. A low-cost monitoring system with continuous logging and prompt alerts protects stock, reduces wastage and provides records for audits.

C · CORE FEATURES

- Measure temperature at one or more points inside the storage unit at a defined interval.
- Detect excursions above and below the allowed range, considering both magnitude and duration.
- Detect door-open events and power failures, and indicate them in the log.
- Raise local and remote alerts with escalation if no one acknowledges.
- Store tamper-evident records and generate summary reports for audits.

D · DESIGN REQUIREMENTS

- Users: primary health centres, hospitals, blood banks, pharmacies and vaccine distribution points.
- Cost and simplicity: affordable per unit, easy to install without altering the refrigerator, usable by non-technical staff.
- Reliability: accurate sensing (e.g. within ±0.5 °C), battery backup during power cuts, few false alarms from routine door openings.
- Compliance: records suitable for immunisation-programme and quality-audit requirements.

E · TECHNICAL FOCUS

SENSING Digital or analogue temperature sensor(s), with probe placement representing the stored product (e.g. in a thermal buffer); optional door switch and mains-presence sensing.

PROCESSING & LOGIC Microcontroller with periodic sampling, filtering, threshold-plus-duration alarm rules and a real-time clock; justify logging interval and alarm delay.

COMMUNICATION Buzzer and display locally, with optional SMS, Wi-Fi, LoRa or app notifications.

POWER SUPPLY Mains with battery backup; state the backup duration.

SIMULATION Thermal model of a storage unit with door openings, compressor cycling and a power outage in Python, MATLAB or equivalent; show alarm behaviour.

OPTIONAL Cloud data logging, multi-unit dashboard, predicted time-to-excursion during outages, calibration reminders.

F · EXPECTED OUTCOME

- System architecture and sensor placement plan.
- Simulation of normal cycling, door-open events, compressor failure and power outage with alarm times.
- Alarm logic with justified thresholds and delays, and its effect on false alarms.
- Sample audit report generated from the simulated data.
- Estimated cost per unit and a note on scaling to many facilities.

PS-02 INDOOR NAVIGATION & OBSTACLE DETECTION AID FOR VISUALLY IMPAIRED PEOPLE

Recommended for First & Second Year students

A · PROBLEM DEFINITION

A white cane detects obstacles near the ground within arm's reach, but it can miss head- and chest-level hazards, overhanging objects, glass doors and drop-offs, and gives no indication of where the user is inside a building. The challenge is to design a low-cost aid that detects obstacles at several heights and directions, communicates them through non-visual feedback and provides simple indoor guidance.

B · CONTEXT

Many visually impaired people in India cannot afford commercial electronic travel aids, which cost far more than a basic cane. Indoors, hazards such as open cupboard doors, furniture edges, stairs and hanging objects are common, and unfamiliar buildings are hard to navigate. A simple device worn or attached to the cane can give timely warnings and basic directional cues, improving safety and independence without advanced technology skills.

C · CORE FEATURES

- Detect obstacles in front, to the sides and at head height, within a useful range.
- Detect stairs or drop-offs using downward-facing sensing.
- Provide graded feedback, e.g. vibration intensity or audio pulse rate that changes with distance.
- Offer simple indoor guidance using pre-placed tags or beacons with spoken or tone cues (room or corridor identification).
- Warn the user of low battery and allow an easy on/off and volume or intensity setting.

D · DESIGN REQUIREMENTS

- Users: visually impaired students, workers and elderly people, with input from users and trainers.
- Affordability: much cheaper than commercial aids, with a justified cost target.
- Usability: light, comfortable, quick to learn, a full day of battery life; it complements the cane rather than replacing it.
- Safety: comfortable vibration and sound levels, reliable alerts and clear behaviour when a sensor fails.

E · TECHNICAL FOCUS

SENSING Ultrasonic and/or infrared time-of-flight sensors, mounted to cover different heights and directions. Discuss limitations such as soft surfaces, glass, sunlight and angled walls.

PROCESSING & FEEDBACK Microcontroller mapping distance to vibration and/or sound patterns; latency low enough to give reaction time at a walking speed of about 1 m/s.

NAVIGATION BLE beacons, RFID or NFC tags at key locations with simple audio prompts; no mapping or complex localisation required.

POWER SUPPLY Rechargeable battery with a stated runtime and low-battery alert.

SIMULATION 2D floor-plan simulation in Python, MATLAB or equivalent with sensor field-of-view models, obstacles and a walking user; report detection distance and warning time.

OPTIONAL Voice assistant, smartphone companion app, camera-based object recognition, wrist-worn feedback unit.

F · EXPECTED OUTCOME

- Block diagram, sensor layout (height and angle) and feedback scheme.
- Simulation of walking through a room with several obstacle types, showing warning distance and reaction time.
- Evaluation of sensor limitations and how the design compensates for them.
- Simple indoor-guidance demonstration using tags or beacons on a sample floor plan.
- Cost estimate and suggestions for user testing in a later stage.

PS-03 BACKUP POWER MONITORING & FAILURE ALERT FOR CRITICAL HOSPITAL EQUIPMENT

Recommended for Second & Third Year students

A · PROBLEM DEFINITION

Intensive-care units, operating theatres, ventilators, incubators and dialysis machines depend on a chain of backup power: UPS, batteries, generators and automatic transfer switches. A weak link in that chain, such as a degraded battery or a generator that fails to start, is often discovered only during an actual outage. The challenge is to monitor the health and readiness of the backup chain and to warn staff early.

B · CONTEXT

Power failures in hospitals can be life-threatening, and routine checks usually test only whether equipment turns on, not how long it will last under load. Battery ageing, loose connections, overloaded UPS units and fuel shortages stay hidden. A monitoring system that tracks mains, UPS and generator status, estimates remaining runtime and escalates alerts can reveal weaknesses in advance and shorten response time during a failure.

C · CORE FEATURES

- Monitor mains presence, UPS/battery voltage, current, temperature and load level.
- Estimate remaining backup runtime at the present load and flag when it falls below a safe limit.
- Detect generator start failure, transfer-switch failure and slow changeover.
- Identify degrading battery health from indicators such as voltage sag under load and temperature rise.
- Escalate alerts to the right people (nurse station, biomedical engineer) and keep an event log.

D · DESIGN REQUIREMENTS

- Users: hospitals, nursing homes, diagnostic centres and clinics.
- Reliability: the monitor itself must be highly dependable, with self-checks, a watchdog and its own backup power, and must not interfere with the equipment it monitors.
- Standards & practice: consistent with hospital accreditation and electrical safety; justify changeover targets (no break for UPS, a few seconds for a generator).
- Cost & integration: affordable retrofit on existing backup systems, optional link to a building-management system.

E · TECHNICAL FOCUS

SENSING Voltage and current sensing on mains, UPS output and battery strings; temperature sensors; generator and transfer-switch status contacts; optional fuel-level sensing.

ANALYTICS Runtime estimation from state of charge and load; health indicators from voltage drop and internal-resistance estimates; alert-severity rules.

PROCESSING & COMMUNICATION Microcontroller or industrial controller; Modbus/RS-485, Wi-Fi or GSM links; alert escalation with acknowledgement tracking.

POWER SUPPLY Independent supply with its own battery so monitoring survives a total outage.

SIMULATION Simulate mains failure, UPS discharge, generator start success and failure and battery degradation in Simulink, Python or equivalent; show alerts and estimates.

OPTIONAL Predictive battery-replacement planning, load-shedding advisory, integration with a nurse-call or building system.

F · EXPECTED OUTCOME

- System architecture covering sensors, controller, communication and alert paths.
- Simulation of at least four events (mains loss, generator success, generator failure, degraded battery) with detection times.
- Runtime-estimation method and its accuracy against the simulated discharge.
- Alert-escalation scheme with example messages.
- Cost estimate and discussion of the monitor's own fail-safe design.

PS-04 REAL-TIME LOCATION & AVAILABILITY MONITORING OF SHARED HOSPITAL EQUIPMENT

Recommended for Second & Third Year students

A · PROBLEM DEFINITION

Hospitals share movable equipment such as wheelchairs, infusion pumps, ECG machines and portable monitors across many wards. Staff spend valuable time searching for an item, and nobody knows whether it is free, in use, awaiting cleaning or due for maintenance. The challenge is to track the location and status of shared equipment in near real time and to help staff find an available item quickly.

B · CONTEXT

Time lost searching for equipment delays care, while hospitals over-purchase to compensate for items that are simply hard to find. Manual registers are rarely updated, and tracking systems based on expensive infrastructure are out of reach for many hospitals. A low-cost tag-based system that provides room- or zone-level location together with a status indicator can improve responsiveness, equipment utilisation and maintenance scheduling.

C · CORE FEATURES

- Attach tags to equipment and determine location at room or zone level.
- Track status: available, in use, in cleaning, under maintenance or missing.
- Provide a search interface that shows the nearest available item of a given type.
- Alert when equipment leaves its authorised zone, when a tag battery is low or when maintenance is due.
- Log usage and produce utilisation reports.

D · DESIGN REQUIREMENTS

- Users: hospitals of different sizes, with nurses, porters and biomedical staff as day-to-day users.
- Cost: low cost per tag and per fixed receiver, long tag battery life (target one year or more).
- Hospital constraints: low-power radio that does not interfere with medical devices, tags that withstand cleaning and disinfection, easy attachment.
- Privacy & integration: track equipment only, not patients or staff; ability to integrate with hospital information systems.

E · TECHNICAL FOCUS

POSITIONING TECHNOLOGY BLE, RFID, UWB, Wi-Fi or hybrid options compared on accuracy, cost, battery life and infrastructure; justify the choice.

LOCATION ALGORITHMS Nearest-receiver, RSSI-based trilateration or fingerprinting; discuss the effect of walls, people and metal on signals.

STATUS SENSING Simple tag-side inputs such as motion detection or a push button/state indicator; software inference of use from movement.

SOFTWARE & COMMUNICATION Gateways, database and a staff-facing web or mobile interface with search and alert functions.

SIMULATION Floor-plan simulation with receivers and moving tags using a path-loss model with noise in Python or MATLAB; report location accuracy and misclassification rate.

OPTIONAL Sensor-based status detection (e.g. pump running), integration with cleaning workflows, heat maps of equipment flow, mobile staff app.

F · EXPECTED OUTCOME

- System architecture and a receiver/tag placement plan for a sample ward.
- Simulation showing room-level accuracy (%) and its dependence on noise and receiver density.
- Status model with an example search and alert workflow.
- Estimated infrastructure and tag cost for a 200-bed hospital.
- Discussion of limitations, privacy and interference.

PS-05 AFFORDABLE HAND REHABILITATION DEVICE WITH MOVEMENT ASSISTANCE & FEEDBACK

Recommended for Third & Final Year students

— A · PROBLEM DEFINITION

Stroke, spinal cord injury, nerve damage and arthritis can weaken or paralyse the hand, making grasping and fine tasks difficult. Recovery depends on many repetitions of guided finger movement, which are limited by the shortage of therapists and by the high price of robotic gloves. The challenge is to design an affordable, safe device that assists finger movement, measures progress and gives feedback to the patient and therapist.

— B · CONTEXT

Rehabilitation exercises work best when they are frequent, correctly performed and monitored, but most patients can attend therapy only a few times a week. Commercial robotic rehabilitation gloves cost many lakhs of rupees, out of reach for most patients and clinics. A low-cost device that can assist, resist or guide finger motion, record range of motion and effort, and support home use under remote supervision could widen access to effective therapy.

— C · CORE FEATURES

- Assist finger flexion and extension through a safe actuation method (tendon, linear or soft pneumatic).
- Measure joint angles and movement quality using flex sensors, IMUs or similar.
- Support multiple modes: passive guided movement, active-assist (triggered by patient effort) and resistive exercise.
- Limit force, speed and range of motion for safety, with an emergency stop and fault detection.
- Give visual or audio feedback and record sessions to show progress to therapists.

— D · DESIGN REQUIREMENTS

- Users: stroke and injury patients, therapists, rehabilitation centres and home-care settings.
- Affordability: a cost target that is a small fraction of commercial devices.
- Safety & comfort: safe limits on force and temperature, skin-friendly materials, adjustability for different hand sizes; spasticity must be considered.
- Regulatory & ethical context: medical-device requirements apply to later stages; no testing on human subjects is required or expected in Round 1.

— E · TECHNICAL FOCUS

BIOMECHANICS Finger kinematics (e.g. a three-segment finger model) with typical ranges of motion; estimate the required joint torque and actuator force.

ACTUATION Servo, linear actuator, shape-memory or pneumatic options compared on force, speed, weight, noise and cost.

SENSING & CONTROL Flex sensors, IMUs and optional surface EMG; control strategies such as position, impedance/admittance or EMG-triggered assistance with safety limits.

ELECTRONICS & SOFTWARE Microcontroller, motor drivers, battery power and a simple app or screen for exercises, feedback and session data.

SIMULATION Kinematic and dynamic finger model with the actuator and controller in MATLAB/Simulink, Python or equivalent; use synthetic or published EMG data for the triggered mode.

OPTIONAL Gamified exercises, tele-rehabilitation link, adaptive difficulty from progress data, 3D-printed adjustable chassis.

— F · EXPECTED OUTCOME

- System architecture covering mechanics, sensing, control and user interface.
- Simulation of finger motion for guided, active-assist and resistive modes, with joint-angle tracking error.
- Safety analysis showing force and range limits and behaviour under fault or spasm.
- Demonstration of progress metrics (range of motion, repetitions, effort) from simulated sessions.
- Bill of materials, estimated cost and a comparison with commercial rehabilitation devices.